24 Here are the first 7 terms of a quadratic sequence.

3 6 11 18 27 38 51

(a) Find the next term in this sequence.

Differences: 3, 5, 7, 9, 11, 13; next15.

Next term = 51 + 15 = 66.

1 mark
66

(1)

The nth term of a different sequence is $2n^2 + 5$

(b) Work out the 6th term of this sequence.

For $n = 6$: $2 \times 6^2 + 5$

$= 2 \times 36 + 5 = 77.$

1 mark
77

(1)

(Total for Question 24 is 2 marks)